Puzzles with rules

Optimization · Level 1 · Lesson 3



In this lesson you will

- Solve a 4 × 4 sudoku with its rules.
- Cross out what can’t go in a square.
- Guess only when you’re stuck, and undo wrong guesses.

Key ideas

- No number twice in a row, column or box
- Only one candidate left? Write it in
- Stuck? Guess where there are fewest
- Rules first: 4 guesses, not 13,528

Activity: Sudoku swap

1. Fill (6 min): Fill a 4 × 4 grid with 1 to 4.
2. Hide (3 min): Copy only five or six numbers.
3. Solve (8 min): Swap and solve by crossing out.
4. Share (3 min): Who needed a guess?

Use a pencil so you can rub out a wrong guess. No device needed.

Let’s discuss

1. Why did crossing out save so many guesses?
2. What do you do at a dead end?
3. Which fair rule would you add to your school timetable?

Exit ticket

- One rule of sudoku + what I do when I’m stuck.